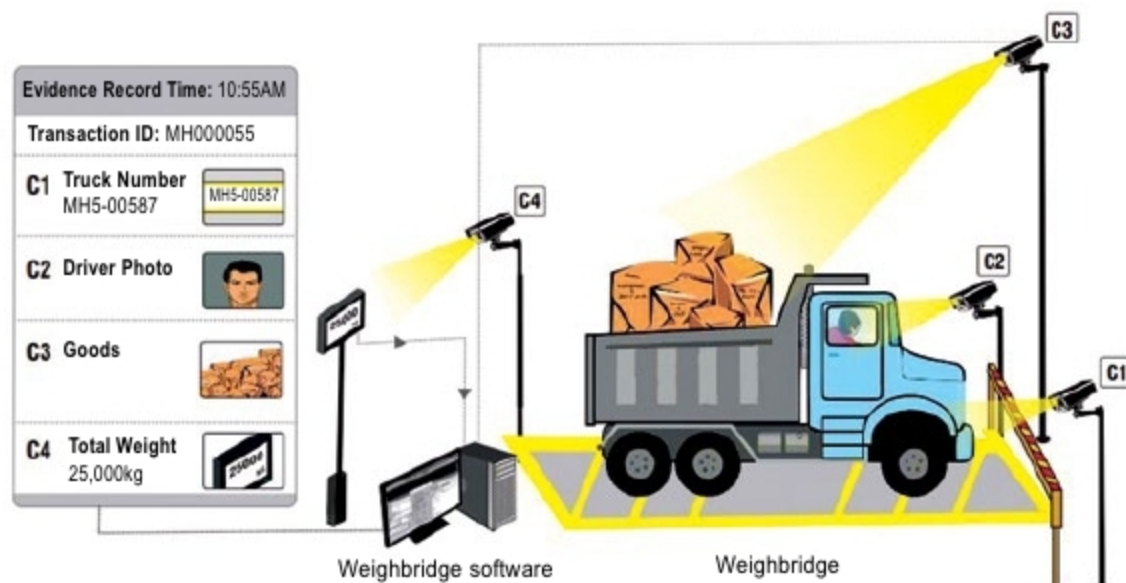


4 AUTOMATION IN LOGISTICS

HELPING PREVENT FRAUD

Weighbridge solutions are an important investment when it comes to the success of operations at logistics firms



For industries such as those related to metal and mining, heavy vehicles are used to transport materials from one place to another. Both dispatch of products from one end and receiving on the other need to be hassle-free. For this, transported goods must reach the destination on time without any pilferage (especially loose materials that cannot be packed) to avoid monetary losses and maintain trust of customers.

To ensure transparency and reliability, weighbridge monitoring solutions provide visual details of the complete loading process to verify that the material and its quantity are the same at both source and destination.

Many such companies as Imagic Solution and Suraj Informatics provide weighbridge solutions. Apart from equipment for the automated system, these companies offer numerous weighbridge data management software that work in synchronisation. Software have functions for data extraction, collection and processing, and can be customised as per requirements.

The admin/user can set different parameters like minimum weight limit and time for stable weight, and reports are generated without an operator. For security, some businesses also provide fingerprint login option.

Companies like Glocon provide vehicle positioning systems so that

vehicles are accurately rested during weighing. For this, photosensors are used to calculate the degree of error; a green light denotes the right position for weight measurement.

A number of companies worldwide have employed integrated weighbridge and software systems in the past few years. A manufacturing company in Kutch, India, chose Satatya Samas weighbridge integration solution by Matrix Comsec as an alternative to its pre-existing solution. The conventional weighbridge system used earlier was suitable for executing basic functionalities like weighing vehicles or containers, and required operator assistance to a great extent.

In this solution, such problems as errors caused by manual interruptions and data spoofing (mostly intentional) were prominent. The manual data entry process was complex, and there was no evidence or detailed information, which is necessary for auditing.

In the new optimised solution, Satatya CIBR20VL12CWS (a 2MP varifocal IR bullet camera with 2.8mm - 12mm lens) and Satatya CIBR20FL36CWS (a 2MP fixed lens IR bullet camera with 3.6mm lens) were selected for surveillance. Here, when a vehicle comes to the weighing scale, one camera focuses on the licence plate of the vehicle, while another overhead camera captures images of the con-

tainer. The third camera is used to take a snapshot of the weighing scale.

Integration with the video management system software consisting of inbuilt weighbridge application module provides ease of managing data from a single system. Information is sent to the system, and can also be printed for proof. The entire process is repeated after loading the vehicle with goods. A supervisor audits, and a manager approves all transactions after checking the images.

After dispatch, this information can be sent to the customer via email and SMS. The option of generating periodic analytical reports of the entire transaction containing information about transaction conditions and other customised fields is helpful for businesses.

In Malaysia, a steel-mill switched to a secure unmanned weighbridge system by Avery Malaysia when inaccuracies in its pre-existing system started to pose problems. In addition to huge production volumes, there was no centralised management system for multiple branches. The existing system was not replaced; instead, it was converted to meet new requirements by adding new load cells and weight indicators. Apart from software, the CCTV system now enables monitoring of every transaction, and snapshots provide evidence in case of disputes, as per www.averyweigh-tronix.com

Automation in internal processes eliminates the need for an operator, reduces errors, increases productivity and results in huge annual savings for businesses. Greater security decreases the possibility of false entries and ensures that every employee obeys the rules set by the organisation.

Weighbridge solutions are an important investment when it comes to the success of operations at logistics firms. With smart automation, such businesses can aim for higher revenues and improved efficiencies. **EFY**

—Ayushee Sharma